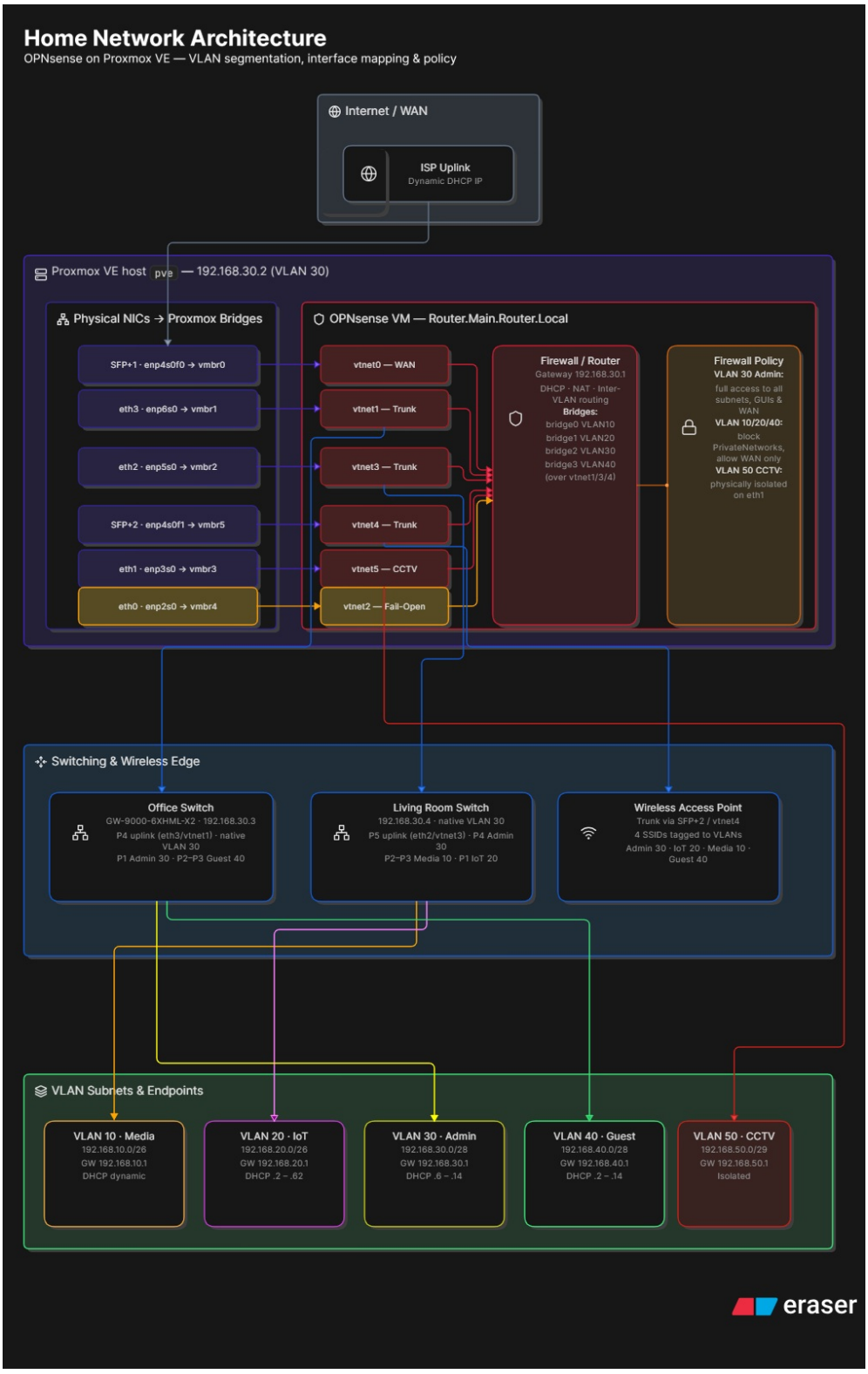
Home Network Architecture

OPNsense on Proxmox VE — VLAN Segmentation, Interface Mapping & Policy



1. Executive Summary

This document describes the complete network architecture for a home network environment utilizing OPNsense firewall virtualized on Proxmox VE. The design implements VLAN segmentation across five network zones, each with a dedicated security policy.

2. Core System Architecture & Management

2.1 Primary Components

Component	Role	IP Address	Location
OPNsense	Firewall / Router	192.168.30.1	Proxmox VM (Router.Main.Router.Local)
Proxmox VE	Hypervisor	192.168.30.2	Physical Host (pve)
Fail-Open Port	Emergency Access	192.168.1.0/24	eth0 → vmbr4 → vtnet2

2.2 Design Principles

- **Virtualized Firewall** — OPNsense runs as a VM within Proxmox VE, providing flexibility and snapshot capabilities.
- **VLAN Segmentation** — Five distinct network zones with inter-VLAN routing controlled by firewall policy.
- **Physical Isolation** — CCTV network operates on a dedicated physical interface for complete separation.
- **Fail-Open Access** — Emergency out-of-band management port bypasses primary network infrastructure.

3. Physical-to-Virtual Interface Mapping

Physical If.	Linux Port	Proxmox Bridge	OPNsense If.	Function
eth0	enp2s0	vmbr4	vtnet2	Fail-Open Access (192.168.1.0/24)
eth1	enp3s0	vmbr3	vtnet5	CCTV Network (VLAN 50)
eth2	enp5s0	vmbr2	vtnet3	Living Room Switch Trunk
eth3	enp6s0	vmbr1	vtnet1	Office Switch Trunk
SFP+1	enp4s0f0	vmbr0	vtnet0	WAN / ISP (Dynamic DHCP)
SFP+2	enp4s0f1	vmbr5	vtnet4	Access Point Trunk

4. Network Subnet & Addressing Topology

4.1 VLAN Summary Table

VLAN	Name	Subnet	Mask	Gateway	DHCP Range
10	Media	192.168.10.0/26	255.255.255.192	192.168.10.1	Dynamic
20	IoT	192.168.20.0/26	255.255.255.192	192.168.20.1	.2 – .62
30	Admin	192.168.30.0/28	255.255.255.240	192.168.30.1	.6 – .14
40	Guest	192.168.40.0/28	255.255.255.240	192.168.40.1	.2 – .14
50	CCTV	192.168.50.0/29	255.255.255.248	192.168.50.1	.2 – .6

4.2 Subnet Capacity

- **VLAN 10 / 20** (/26) — 62 usable host addresses each
- **VLAN 30 / 40** (/28) — 14 usable host addresses each
- **VLAN 50** (/29) — 6 usable host addresses

5. OPNsense Bridge Infrastructure

OPNsense utilizes internal bridges to aggregate VLAN traffic across multiple trunk interfaces:

Bridge	VLAN	Member Interfaces
bridge0	VLAN 10 (Media)	vtnet1, vtnet3, vtnet4
bridge1	VLAN 20 (IoT)	vtnet1, vtnet3, vtnet4
bridge2	VLAN 30 (Admin)	vtnet1, vtnet3, vtnet4
bridge3	VLAN 40 (Guest)	vtnet1, vtnet3, vtnet4

6. Switching & Wireless Edge Configuration

6.1 Office Switch (GW-9000-6XHML-X2)

Management IP **192.168.30.3** | Subnet **255.255.255.240** | Gateway **192.168.30.1** | Native VLAN **30 (Admin)**

Port	Assignment
Port 4	Uplink to Firewall (eth3 / vtnet1)
Port 1	Admin VLAN 30 (Untagged)
Ports 2–3	Guest VLAN 40 (Untagged)

6.2 Living Room Switch

Management IP **192.168.30.4** | Native VLAN **30 (Admin)**

Port	Assignment
Port 5	Uplink to Firewall (eth2 / vtnet3)
Port 4	Admin VLAN 30 (Untagged)
Ports 2–3	Media VLAN 10 (Untagged)
Port 1	IoT VLAN 20 (Untagged)

6.3 Wireless Access Point

Uplink **SFP+2 (vtnet4)** | Configuration: trunk port carrying all VLANs

SSID	VLAN Tag
Admin WiFi	VLAN 30
IoT WiFi	VLAN 20
Media WiFi	VLAN 10
Guest WiFi	VLAN 40

7. Security & Firewall Policy

7.1 Policy Matrix

Source Network	Destination	Policy
VLAN 30 (Admin)	All Internal Subnets	✓ Allow
VLAN 30 (Admin)	Management GUIs	✓ Allow
VLAN 30 (Admin)	Internet (WAN)	✓ Allow
VLAN 10 / 20 / 40	PrivateNetworks (RFC 1918)	✗ Block
VLAN 10 / 20 / 40	Internet (WAN)	✓ Allow
VLAN 50 (CCTV)	All Networks	✗ Isolated

7.2 Policy Details

Admin Network (VLAN 30)

- Unrestricted outbound access to all internal subnets
- Full access to OPNsense GUI (192.168.30.1)
- Full access to Proxmox VE GUI (192.168.30.2)
- Full access to switch management (192.168.30.3, 192.168.30.4)
- Unrestricted Internet access

Media, IoT & Guest Networks (VLANs 10, 20, 40)

- Explicit block rule against the PrivateNetworks alias (RFC 1918)
- Allow rule for outbound WAN access only
- No inter-VLAN communication permitted

CCTV Network (VLAN 50)

- Physically isolated on a dedicated interface (eth1 / vtnet5)
- Complete separation from all client networks
- No routing to other VLANs